

IMO Training 2020–21 (Phase 1)
 Level 1
 Introduction to IMO and Problem Solving Strategies

Andy Loo
 andyloo1994@gmail.com

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1 Introduction to IMO training and competitions

Activity	Eligibility	Time	Format	Location	Difficulty
IMO HK Prelim	6 per school + previous year's awardees	May	3 hours, 20 answer-only questions	Usually La Salle College	★
Phase 1 training	Prelim awardees and more	Jul – Aug	Lessons	HKAGE	NA
Test 1	Phase 1 trainees	Aug	3 hours, 6 proof problems	HKAGE	★ ★
Phase 2 training	Same as Phase 1	Sep – Dec	Lessons	HKAGE	NA
Test 2	Phase 2 trainees	Around Oct	3 hours, 4 proof problems	HKAGE	★ ★
CHKMO	Phase 2 trainees	Dec	3 hours, 4 proof problems	HKAGE	★ ★ ★
CMO	Team of 6, based on Prelim, Test 1, Test 2	Dec or Jan	2×4.5 hours, 2×3 proof problems	A city in China	★ ★ ★ ★ ★ ★

Activity	Eligibility	Time	Format	Location	Difficulty
Phase 3 training	Same as Phase 1	Jan – Mar	Lessons	HKAGE	NA
APMO	Phase 3 trainees	Mar	4 hours, 5 proof problems	HKAGE	★ ★ ★ ★
Phase 4 training	IMO team & alternate members	Around Jun – Jul	Lessons & mock tests	HKAGE	NA
IMO	Team of 6, based on Prelim, Test 1, Test 2, CHKMO, APMO, further tests	Jul*	2×4.5 hours, 2×3 proof problems	2021: TBC	★ ★ ★ ★ ★
CWMI	Team of 8, F.5 or below, based on Prelim, Test 1, Test 2, CHKMO, APMO	Aug	2×4 hours, 2×4 proof problems	A city in western China	★ ★ ★ ★
CGMO	Team of 8 girls, via CGMO training (NOT IMO training)	Aug	2×4 hours, 2×4 proof problems	A city in China	★ ★ ★ ★

* Due to COVID 19, IMO 2020 is postponed to September and will be held virtually.

Other learning resources:

- We will provide details on joining the IMO Training Google Drive folder soon. Please prepare a Google account in the meantime.
- IMO official website: www.imo-official.org
- Art of Problem Solving: www.artofproblemsolving.com
- Mathematical Excalibur: www.math.ust.hk/excalibur/

2 Problem solving strategies

2.1 Reducing to known problems

Problem 1 Let D be a point inside $\triangle ABC$ (possibly on its boundary). Prove that

$$AB + AC \geq DB + DC.$$

Problem 2 Prove that among any 6 people in the world, 3 of them know each other or 3 of them do not know each other. (Knowing is mutual.)

Problem 3 (IMO 1964) 17 people correspond by mail with one another – each one with all the rest. In their letters only 3 different topics are discussed. Each pair of correspondents deals with only one of these topics. Prove that there are at least 3 people who write to each other about the same topic.

2.2 Finding patterns

Problem 4 (IMO HK Prelim 2009) Let $\{x_n\}$ be a sequence of positive real numbers. If $x_1 = \frac{3}{2}$ and $x_{n+1}^2 - x_n^2 = \frac{1}{(n+2)^2} - \frac{1}{n^2}$ for all positive integers n , find $x_1 + x_2 + \dots + x_{2009}$.

Problem 5 (CWMO 2010) Suppose n players participate in a table tennis tournament ($n \geq 2$). Every two players play against each other exactly once. A player A is said to be *no worse than* another player B if either (i) A beats B or (ii) A beats someone who beats B . Find all possible values of n such that the following case is possible: no player is worse than any other player.

2.3 Combining numbers and figures

Problem 6 Two secret agents, Kelvin and James, are to meet at a park today, but they don't know when each other will arrive. Each person independently, randomly and uniformly picks an arrival time between 5:00 pm and 6:00 pm. Each of them will leave after waiting for 10 minutes to avoid danger. Find the probability that they will successfully meet at the park.

Problem 7 Prove that for any positive real numbers x , y and z , we have

$$\sqrt{x^2 - xz + z^2} + \sqrt{y^2 - yz + z^2} \geq \sqrt{x^2 + xy + y^2}.$$

2.4 A final example

Problem 8 Let $ABCD$ be a square with side length 1. If E is a point inside the square, find the smallest possible value of $AE + BE + CE$.